

IB MYP
Formative Assessment - 1 (2026-27)

Students Name:	Grade & Sec: MYP-1	Date: -07-2026
Subject: Integrated Sciences	Task Duration: 1 hour	Level Obtained
Unit Title : What do Scientists do?		
Criteria & Strands assessed: CRITERION C: Processing and Evaluating ● Present collected and transformed data ● Interpret data and explain results using scientific reasoning ● Discuss the validity of a hypothesis based on the outcome of the scientific investigation ● Discuss the validity of the method ● Describe improvements or extensions to the method.		

CRITERION-C : Processing and Evaluating

Achievement Level	Level Descriptor	Task-Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	The student does not reach a standard
1-2	The student should be able to: i. collect and present data ii. interpret data iii. state validity of prediction (limited success) iv. state validity of method (limited success) v. state improvements (limited success)	The student is able to: i. Collect and present data in numerical and/or visual forms. ii. Interpret the data about iii. State whether the prediction is supported by the results. iv. State whether the investigation method was suitable, with limited success. v. State one improvement or extension to the investigation with limited success.
3-4	The student should be able to: i. correctly collect and present data ii. accurately interpret data and outline results iii. state validity of prediction iv. state validity of method v. state improvements	The student is able to: i. Correctly collect and present data in numerical and/or visual forms ii. Accurately interpret the data and outline the results of the investigation. iii. State whether the prediction is valid based on the investigation results. iv. State whether the investigation method was valid. v. State improvements or extensions that

		would improve the investigation.
5-6	The student should be able to: i. correctly collect, organize and present data ii. interpret data using scientific reasoning iii. outline validity of prediction iv. outline validity of method v. outline improvements	The student is able to: i. Correctly collect, organize and present data in numerical and/or visual forms. ii. Interpret the data and explain the results using scientific reasoning. iii. Outline whether the prediction is valid based on the investigation results. iv. Outline whether the investigation method was valid. v. Outline improvements or extensions that would improve the investigation.
7-8	The student should be able to: i. correctly collect, organize, transform and present data ii. interpret data using correct scientific reasoning iii. discuss validity of prediction iv. discuss validity of method v. describe improvements	The student is able to: i. Correctly collect, organize, transform and present data in numerical and/or visual forms. ii. Interpret the data and explain the results using correct scientific reasoning. iii. Discuss whether the prediction is valid based on the investigation results. iv. Discuss whether the investigation method was valid. v. Describe improvements or extensions that would improve the investigation.

Task-1: Your school is planning to purchase paper towels for all science laboratories and washrooms. Before making a decision, the management wants evidence about which brand absorbs the most water. Your class acted as a team of scientists to investigate this question.

Now that the experiment has been completed, your responsibility is to analyse and evaluate your investigation just as professional scientists would. Use your experimental data to justify every conclusion you make.

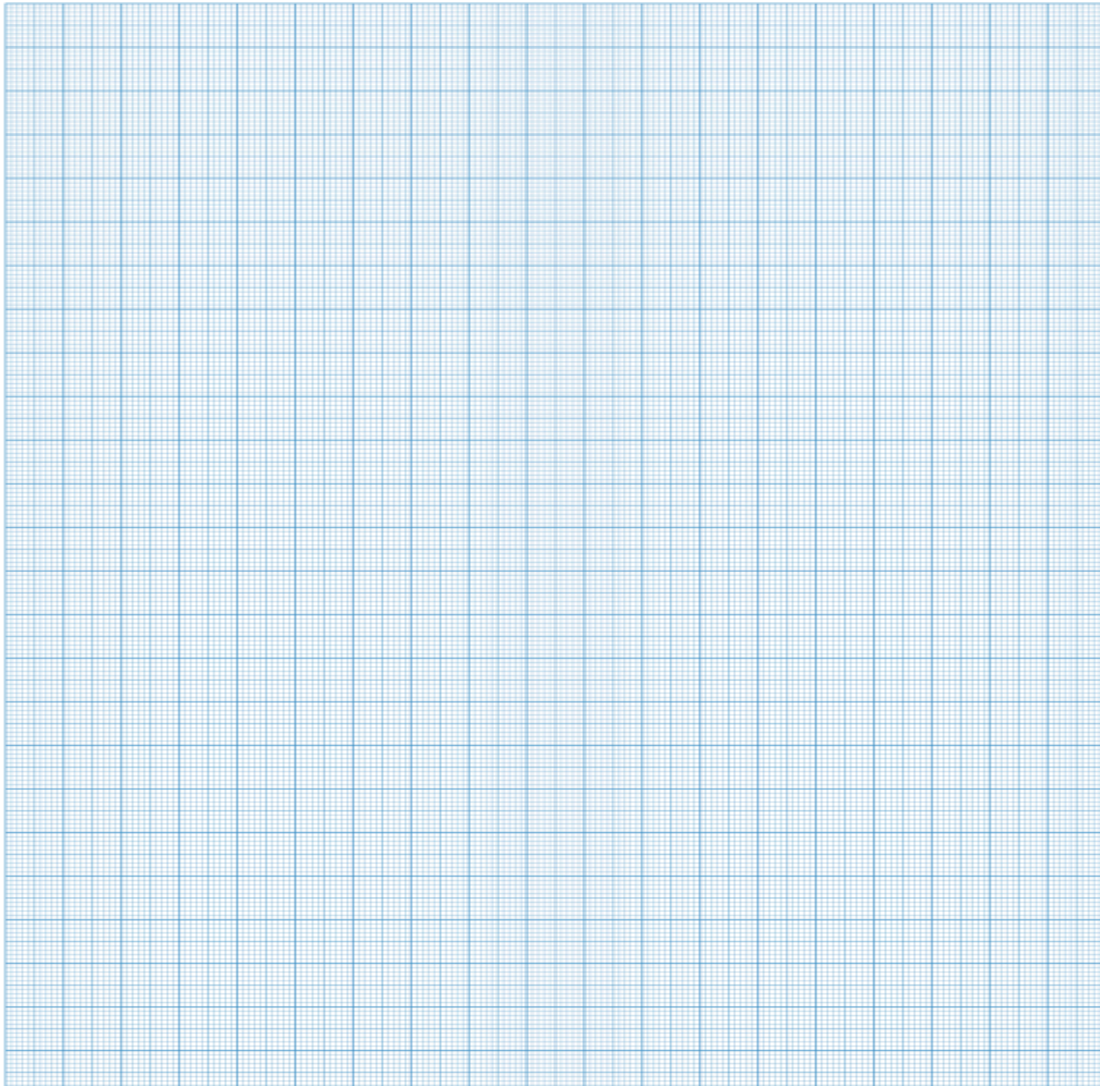
Strand i: Presenting Data

Table 1: Raw Data – Water Absorbed by Different Paper Towel Brands

	Trial-1 (mL)	Trial-2 (mL)	Trial-3 (mL)	Average/mean
Brand A	50	52	51	
Brand B	44	46	45	
Brand c	58	60	59	

Present and transform the data provided by:

- calculating the **average amount of water absorbed** for each paper towel brand,
- plotting a **bar graph** using the average water absorbed,
- labelling both axes with appropriate units,
- choosing a suitable scale, and
- giving the graph an appropriate title.



Strand ii: Interpret Data and Outline Results

Interpret the data presented in your data table and bar graph.

Describe the relationship between the paper towel brand and the amount of water absorbed. Identify any patterns or trends you observed and support your answer with evidence from your data.

Strand iii: Discuss the Validity of a Prediction

Prediction:

"If different brands of paper towels are tested, then Brand C will absorb the greatest amount of water because it is designed to be thicker and more absorbent."

Discuss the validity of this prediction using evidence from the data collected during the investigation. State whether the prediction was supported or not and explain your reasoning.

Strand iv: Discuss the Validity of the Method

Consider:

- whether it was a fair test,
- whether all variables were controlled,
- whether the data collected was reliable,
- whether there were any sources of error or limitations.

Strand v: Describe Improvements or Extensions

a) Describe two improvements that would increase the accuracy or reliability of this investigation.

b) Suggest one extension to this investigation.

For example, describe another factor that could be investigated or explain how the investigation could be expanded.
